

CAT: A Consensus-Adaptive Trust Management based on the Group Decision Making in IoVs

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Abstract—Securing Internet of Vehicles (IoVs) systems against common threats such as false message injection remains challenging, and one typical approach is to deploy trust management solutions. In this work, we propose a Consensus-Adaptive Trust management (CAT) based on the Group Decision Making (GDM) in IoVs. Specifically, in our approach the consensus levels are calculated to measure the difference of opinions (trust values) among vehicles. To estimate the reliability of consensus levels, the divergence between consensus levels is calculated, namely: consensus level similarity. GDM allows us to dynamically adjust the opinion of vehicles (namely: Consensus Reaching Process, CRP). Then, a credit guarantee mechanism is designed to improve the efficiency of CRP and seek out malicious vehicles quickly. To empower the adaptability of trust management for changing environments in IoVs, CAT dynamically manages opinions of vehicles, self-confidence, and their consensus thresholds according to the feedback of delivered messages. Extensive simulation results show the potential of CAT operating in high-risk scenarios, and outperforming other competing baseline methods in terms of accuracy, precision, recall, and F-score.

Index Terms—IoVs, Consensus Level, Group Decision Making, Trust Management

I. INTRODUCTION

A broad range of trust management solutions have been designed to improve the reliability and security of Internet of Vehicles (IoVs) systems [1]–[3], although a number of challenges and limitations associated with existing systems remain [4]–[7]. One such challenge is to reliably detect malicious vehicles without relying on the roadside infrastructure [8]. One alternative solution is to utilize group decision making (GDM), which promotes consensus reaching process (CRP) based on opinions of vehicles [9]. In such a setting, vehicles adjust their opinions to reach a consensus on whether a vehicle can be trusted in groups. Unlike consensus in blockchains, the consensus in GDM indicates the divergence of opinions less than a consensus threshold, and CRP iteratively adjusts opinions to reach a consensus [10]. A key difference is that GDM has content (opinions) consistency within a tolerant zone, while strong content consistency exists in blockchains,

i.e., the complete same content [11]. However, one known limitation of GDM is the inefficient iteration process in CRP.

To mitigate the above limitation, GDM calculates consensus levels and their divergences to evaluate the consistency degree [12]. The consensus level indicates the consistency of opinions in a group, and is evaluated according to opinions, the weight¹ of opinions, and consensus thresholds. Recent studies on GDM show that opinion adjustment methods decrease computational overhead and improve their iterative performance [13]–[15]. Although above studies effort to strike a balance between opinion adjustment and iterative counts, they ignore the influence of trustworthiness of opinions. Specifically, untruth opinions provided by malicious vehicles affect the number of iterative rounds of CRP, resulting in a high computational overhead and making false decisions. Therefore, the identification of untruth opinions and malicious vehicles is important for GDM to make the right decision, because the adjustment of opinions depends on their trustworthiness.

In addition, trust management plays a significant role in identifying malicious vehicles and providing reliable opinions [16], to defend against several malicious behaviors, e.g., Message Tampered Attacks (MTA), bad mouth attacks, and collusion attacks [4]. Several trust values are considered to evaluate the trustworthiness of vehicles based on historic interactions, such as data trust, direct trust, and recommend trust [17]–[19]. Meanwhile, recent works on trust management generally rely on the assistance of infrastructures, e.g., roadside units. For instance, infrastructures collect various trust values, and calculate a comprehensive trust value based on a utility function. Then, they utilize a constant trust threshold to identify the trustworthiness of vehicles [20]. However, they integrate various trust values to improve the precision of trust evaluations, but fail to reasonably update trust thresholds to adapt to variable environments (different scales of groups).

Based on the above consideration, to open an avenue for exploring the integration of GDM and trust management, we propose a Consensus-Adaptive Trust management (CAT) based on GDM in IoVs. The main contributions are summarized as follows:

- 1) We first construct a holistic system by integrating GDM and trust management, i.e., CAT, to improve the identification performance of trust models and the adaptability of dynamic environment. Specifically, GDM is utilized to adjust and aggregate trust values based on various

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¹The weight represents the importance of something.

consensus levels, and trust management is employed to manage the trustworthiness of vehicles.

- 2) We propose the concept of self-certifying consensus level to explore a new dimensional indicator for consensus level evaluation. It shifts the view of trustworthiness verification from being passively estimated to forwardly providing, which is benefit for evaluating the divergence of opinions.
- 3) We propose a credit guarantee mechanism to quickly seek out potential malicious vehicles. It expands the range of assessed vehicles based on the guaranteed behavior, i.e., one vehicle (guarantor) vouches for the assessed vehicle (evaluatee).
- 4) To improve the adaptability of trust management in a variable environment, we propose a consensus threshold dynamic adjustment method according to the weight of opinions and the trust tolerance² of vehicles. It alleviates the problem of a constant threshold that cannot dynamically change with a variable environment.

The remainder of this paper is organized as follows. Section II introduces the related works. Section III illustrates the details of a system model. The flow of our proposed algorithm is shown in Section IV. Section V shows the simulation setting and experiment results. Section VI expresses the conclusion of this paper.

II. RELATED WORK

A. Group Decision Making

The CRP in GDM iteratively adjusts the opinion of non-consensus vehicles until their opinions can be acceptable. Specifically, to establish the connectivity between bounded confidence and CRP, the work [21] proposed a personalized feedback-based GDM mechanism with bounded confidence learning, considering the acceptability of vehicles and comprehensive opinions. The work [22] proposed a trust and fuzzy-based framework to build a recommendation mechanism with the limitation of interval-valued information. It introduces an attitude degree to evaluate the weight of vehicles, and a harmony degree is considered to control the difference between the original opinion and the adjusted one. To deal with the problem of individual and group manipulation, the work [23] proposed an optimal feedback model based on the minimum adjustment cost, the weight of vehicles, and consensus thresholds. An opinion boundary is utilized to adjust the opinion of vehicles that have not reached a consensus in groups.

Several data-driven methods of GDM improve the ability of data processing and utilize more information to establish the relationship between vehicles, such as social relationships, linguistic information, and multi-attributes. The work [24] proposed a linguistic form-based GDM to support the revision of opinions according to the behaviors of opinion adoption

and group consensus. It evaluates the similarity concept between two linguistic tuples to calculate the support of group members. The work [25] proposed an automatic consensus reaching method with preference adjustment willingness of vehicles, and its major aim is to ensure the consistency of consensus thresholds. To address the problem of heterogeneous information and diversified attributes in GDM, the work [26] constructed a multi-granulation-based fuzzy decision-making system. The work [10] proposed an Social Trust Networks (STN)-driven consensus decision framework to cope with Multiple Attributes in GDM, namely MAGDM. It provides a two-stage analysis method for constructing the STN based on trust propagation and aggregation, and considers the minimum adjustment method to reach an optimal consensus for decision-making.

Based on above works, traditional methods of GDM focus on facilitating the convergence process of group opinions based on feedback mechanisms and social relationships. However, although they make more efforts to reduce the computation cost or to balance their opinions with a minimum adjustment, traditional methods ignore the damage of malicious opinions (e.g., collusion attacks). Malicious opinions may mislead the judgment of benign vehicles, causing traffic accidents and threatening life safety, especially in IoVs.

B. Trust Management

Trust management focuses on detecting malicious vehicles by evaluating their trustworthiness, to alleviate the influence of their malicious behaviors [27] (e.g., MTA and bad mouth attacks). A Decentralized Trust Management Scheme (DTMS) [18] was proposed to reduce the dependency level of group cooperation. DTMS combines the forwarding evidence and energy consumption rate to calculate the direct trust, and the recommendation trust is utilized to improve the reliability of trust evaluation. The work [28] proposed a hybrid method to assess and manage vehicles based on the historical interaction perspective. It calculates the direct trust, recommended trust, and reputation with the association of a time sliding window and time decay function. The work [29] proposed a context-aware trust management mechanism, to deal with the service-oriented requirements in IoVs. It introduces a logistic regression and recommendation filtering method, to dynamically evaluate the trustworthiness of service providers based on their service behavior patterns. Then, to adapt to sensor networks that are different from IoVs (i.e., network entities are fixed in a certain location and communicate with unchanged neighbors), the work [16] proposed a trust management for unattended wireless sensor networks to improve the security of whole systems. The work [30] designed a multi-behavior level-based hierarchical trust management, to identify malicious behaviors while maintaining the cooperation of nodes.

To maintain communication stability and prevent broadcast storms, the work [31] proposed a multi-layers trust management framework based on blockchain, considering timestamps, communication qualities, direct trusts, and recommendation trusts. The work [32] proposed an Attack-Resistant Trust management scheme (ART), to deal with malicious behaviors

²The trust tolerance indicates a trust threshold to judge the trustworthiness of vehicles. For instance, when the trust tolerance is 0.7, the trust value of a vehicle is 0.8, and the vehicle is trusted. Vehicles may have different values of the trust tolerance.

by integrating trustworthiness, data trust, recommendation trust, and direct trust. ART establishes and manages three dimensions trusts of vehicles according to the interaction evidence extracted from traffic data. A Bayes theory-based trust model [33] was proposed to deduce the trustworthiness of vehicles, namely Implicit Web of Trust in Vehicular ad-hoc networks (IWOT-V). To improve the detection accuracy of malicious vehicles, IWOT-V originates from the page rank technology and calculates local/global trust relationships based on STN. The work [34] proposed a hybrid trust management to identify compromised information and vehicles, considering the distance between vehicles, data trust, direct trust, and entity trust. It is divided into two stages according to the evaluation process of trustworthiness, i.e., the entity trust evaluation stage and data trust evaluation stage.

However, the majority of traditional trust management mechanisms are designed with centralized or half-centralized frameworks. They fails to adapt the distributed framework in IoVs and relies on a threshold to judge the trustworthiness of vehicles. Furthermore, traditional trust management cannot deal with high percentage of malicious vehicles.

C. Motivation

Based the on above concerns, GDM and trust management have the same underlying logic, i.e., change opinions in GDM and trust values in trust management. It is beneficial for the convergence of opinions in a group and decision-making of trust management. On the one hand, with the increase of computational power of vehicles, the distributed framework of GDM can be employed to associate with the decision-making process of message transmission in IoVs. On the other hand, thanks to the opinion change process of GDM, vehicles can generate comprehensive opinions to select trust vehicles, and manage them according to their social trust relationships, interactions (behaviors), and opinions. Therefore, we integrate the advantages of GDM and trust management for message transmission in IoVs, and propose a consensus-adaptive trust management. To overcome the problem of false opinions faced by GDM, a credit guarantee mechanism is designed based on trust management. It can quickly adjust opinions to the right range.

III. SYSTEM MODEL

A. System Architecture

As shown in Fig. 1, there are several vehicles on the road and established connectivities with each other by wireless links based on 802.11p [35]. Vehicles are classified into three types, i.e., the evaluator, guarantors, and the evaluatee. Specifically, to transmit messages to their destinations, the evaluator is responsible for selecting a next-hop (vehicle) according to opinions of itself and its neighbors. The evaluatee is a candidate vehicle for message transmission. Guarantors are the neighbors of an evaluator and provide positive opinions to the evaluator (i.e., to ensure the reliability of the evaluatee).

Furthermore, opinions indicate trustworthiness among vehicles in terms of attributes (e.g., entity trust, data trust, behavior trust), which are expressed in the form of trust values. In other

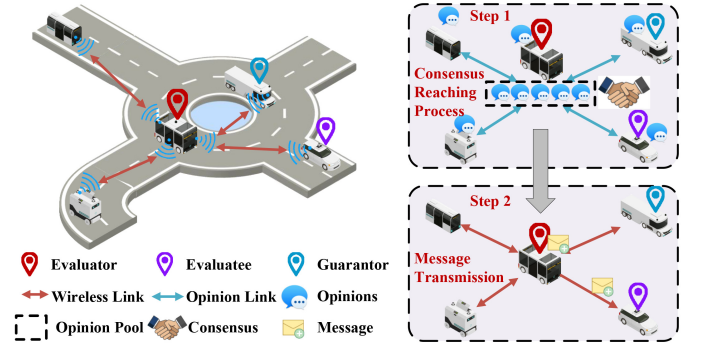


Fig. 1. The scenario of CAT.

words, several attributes comprise an opinion. For instance, $a_{i,k} = \{a_{i,k}^1, \dots, a_{i,k}^j, \dots, a_{i,k}^m\}$ represents the opinion of vehicle v_i on another vehicle v_k , where $a_{i,k}^j$ represents the opinion of j -th attribute on the view of v_i on v_k , and m represents the number of attributes. Moreover, a group consists of several vehicles, and its head (i.e., evaluator) collects opinions from its neighbors (i.e., one-hop vehicles). Based on this, according to received opinions, the evaluator operates CRP by calculating consensus levels (e.g., the subjective, objective, and comprehensive consensus levels), and adjusting opinions (i.e., Step 1 in Fig. 1). Then, a next-hop (i.e., evaluatee) is selected to receive messages (i.e., Step 2). Finally, the detailed process of CAT is shown in Fig. 4 and provides an overview.

B. Adversary Model

Two adversary models are employed, i.e., MTA and Hybrid Attack (HA) [18], [28]. Both MTA and HA are conducted by malicious vehicles. Furthermore, malicious vehicles recognize each other, but benign vehicles do not know. The detail of adversary models is illustrated as follows:

- Message Tampered Attack (MTA): Malicious vehicles tamper with the received messages.
- Hybrid Attack (HA): HA includes bad mouth attacks and collusion attacks. Malicious vehicles provide positive opinions for benign vehicles when the evaluatee is malicious. Similarly, malicious vehicles provide negative opinions for benign vehicles when the evaluatee is benign.

C. Preliminaries

Based on the framework of GDM [36], the decision of message transmission becomes more flexible and adaptive in a trust management system. To clearly understand the underlying logic and relationship between trust management and GDM, the construction of GDM should be introduced firstly. Specifically, a GDM problem is usually modeled as the problem of CRP, i.e., vehicles make a decision from an alternative set according to their opinions [22]. The alternative set is denoted as $V = \{v_1, v_2, \dots, v_n\} (n \geq 2)$, where v_n represents the n -th vehicles.

Furthermore, an opinion matrix is utilized to describe opinions of vehicles on others [21], which is shown as follows:

$$A_i = \begin{bmatrix} a_{i,1}^1 & a_{i,1}^2 & \dots & a_{i,1}^m \\ a_{i,2}^1 & a_{i,2}^2 & \dots & a_{i,2}^m \\ \vdots & \vdots & \ddots & \vdots \\ a_{i,n}^1 & a_{i,n}^2 & \dots & a_{i,n}^m \end{bmatrix}_{n \times m}, \quad (1)$$

where A_i represents opinions of the i -th vehicle, and n represents the number of vehicles, m represents the number of attributes. For instance, $a_{1,2}^m$ represents the m -th opinion of v_1 on v_2 .

In addition, a message feedback-based weight matrix (W_i) is introduced to describe the social relationship of v_i .

$$W_i = \begin{bmatrix} w_{i,1} & w_{i,1}^1 & w_{i,1}^2 & \dots & w_{i,1}^m \\ w_{i,2} & w_{i,2}^1 & w_{i,2}^2 & \dots & w_{i,2}^m \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ w_{i,n} & w_{i,n}^1 & w_{i,n}^2 & \dots & w_{i,n}^m \end{bmatrix}_{n \times (m+1)}, \quad (2)$$

where $w_{i,n}$ represents the weight of v_n on the viewpoint of v_i . $w_{i,n}^m$ represents the m -th weight of v_n on the viewpoint of v_i , which is corresponding to $a_{i,n}^m$.

Based on this, the weight of opinions is dynamically updated according to the number of delivered messages. Then, a message matrix is introduced to update the weight of vehicles (e.g., $w_{i,n}$).

$$D_i = [\{d_{i,1}^s, d_{i,1}^t\}, \{d_{i,2}^s, d_{i,2}^t\}, \dots, \{d_{i,n}^s, d_{i,n}^t\}]_{1 \times n}, \quad (3)$$

where $d_{i,1}^s$ represents the number of non-tampered messages received by v_i passing through v_1 , and $d_{i,1}^t$ represents the total number of transmitted messages through v_1 . Specifically, each message has a list to store the passed vehicles, namely the passed list. If a message is delivered to its destination and is not being tampered with, the value in the message matrix and the weight of passed vehicles will increase. For example, the passed list of a message is denoted as $M_i : \{v_1 \rightarrow v_4 \rightarrow v_7 \rightarrow v_{10}\}$, where v_1 and v_{10} represent the source and destination respectively. v_4 and v_7 are intermittent vehicles, thus $d_{10,4}^s$ and $d_{10,7}^s$ will increase by one. Then, the weight of v_4 and v_7 is updated as show in follows:

$$w_{i,g} = \frac{d_{i,g}^s}{d_{i,g}^t}, \quad (4)$$

where $w_{i,g}$ represents the weight of v_g on the viewpoint of v_i .

D. Consensus Evaluation

Based on the above architecture and preliminaries, the consensus evaluation is employed to estimate the level of consensus among vehicles in a group [37]. It can be divided into four parts from different views, i.e., the subjective consensus level, objective consensus level, comprehensive consensus level, and self-certifying consensus level. Due to an attribute is a small-scale opinion, the subsequent attribute and opinion are jointly referred to as opinion.

Definition 1 Subjective Consensus Level: It indicates the trust value of evaluatee v_k based on evaluator v_i 's opinions, denoted as $SCL_{i,k}$.

$$SCL_{i,k} = \sum_{j=1}^m w_{i,k}^j a_{i,k}^j, \quad (5)$$

where m is the number of opinions, and $a_{i,k}^j$ represents the j -th opinion of v_i on v_k . $w_{i,k}^j$ represents the weight of $a_{i,k}^j$.

Definition 2 Objective Consensus Level: It indicates the trust value of evaluatee v_k based on opinions of v_i 's neighbors, denoted as $OCL_{i,k}$.

$$OCL_{i,k} = \sum_{j=1}^m w_{i,k}^j \sum_{g \in O} \widehat{w_{g,k}^j} a_{g,k}^j, \quad (6)$$

where O represents the neighbors of v_i who has contacted with v_k , and $\widehat{w_{g,k}^j}$ represents the normalized weight according to the weight in set O , i.e., $\widehat{w_{g,k}^j} = \frac{w_{g,k}^j}{\sum_{t \in O} w_{t,k}^j}$.

Definition 3 Comprehensive Consensus Level: It indicates the trust value based on $SCL_{i,k}$ and $OCL_{i,k}$, denoted as $CCL_{i,k}$.

$$CCL_{i,k} = \beta SCL_{i,k} + (1 - \beta) OCL_{i,k}, \quad (7)$$

where β represents the confidence of evaluator v_i .

Definition 4 Self-Certifying Consensus Level: It indicates the trust value of v_k provided by itself, denoted as $SCCL_k$.

$$SCCL_k = \sum_{j=1}^m w_{i,k}^j a_{i,k}^j, \quad (8)$$

where $a_{i,k}^j$ represents the j -th self-certifying opinion. The self-certifying consensus level is utilized to judge whether it is observably distinct from other consensus levels. Specifically, if the similarity between $SCCL_i$ and other consensus levels cannot satisfy a dynamic consensus threshold, v_k has a high probability to operate malicious behaviors.

Definition 5 Consensus Level Similarity: It indicates the divergence between $SCCL_i$ and other consensus levels, denoted as $CLS_{t,k}$, $t \in \{i, o, oi\}$.

$$CLS_{i,k} = 1 - |SCCL_k - SCL_{i,k}|, \quad (9)$$

$$CLS_{o,k} = 1 - |SCCL_k - OCL_{i,k}|, \quad (10)$$

$$CLS_{oi,k} = 1 - |SCCL_k - CCL_{i,k}|, \quad (11)$$

where $t \in \{i, o, oi\}$ is a temporary variable in three consensus levels utilized to calculate the consensus level similarity. Then, the higher the consensus level similarity is, the less divergence of opinions is.

In addition, evaluatee v_k transmits a self-certifying message to evaluator v_i to verify its honesty based on Eq. (8). The structure of self-certifying messages consists of several attributes, i.e., $a_k = [a_k^1, a_k^2, \dots, a_k^m]$. To identify whether a consensus

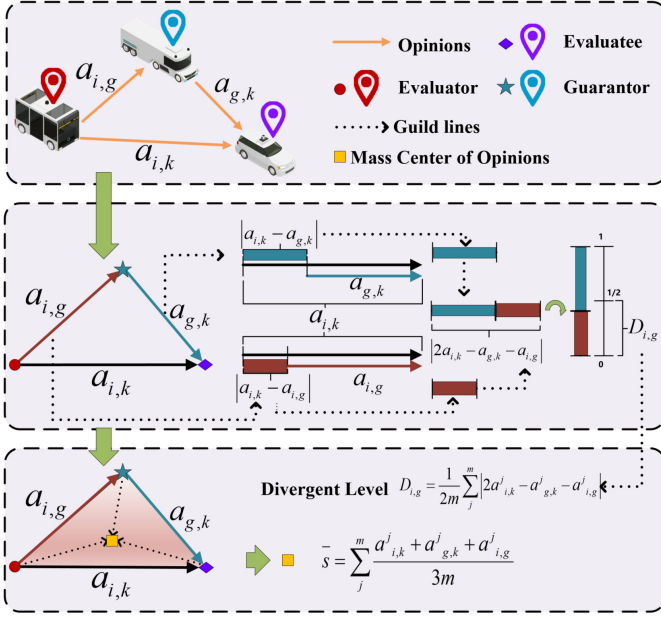


Fig. 2. An example of the credit strength.

level similarity is satisfied, a consensus threshold is employed as shown in follows:

$$CLS_{t,k} \in \{CLS_{i,k}, CLS_{o,k}, CLS_{oi,k}\} \geq \mu, \quad (12)$$

where μ represents the consensus threshold. When the consensus level similarity exceeds μ , it represents that the self-certifying consensus level can be accepted by evaluator v_i . The equation of consensus thresholds is shown as follows:

$$\mu = \beta\mu_i + (1 - \beta) \sum_{g \in O} \widehat{w}_{i,g} \mu_g, \quad (13)$$

where μ_i represents the trust tolerance of v_i , and μ_g represents the trust tolerance of v_g . $\widehat{w}_{i,g}$ represents the normalized weight of v_g on the viewpoint of v_i .

E. Credit Guarantee Mechanism

To seek out potential malicious vehicles and improve the possibility of message transmission, the Credit Guarantee Mechanism (CGM) is designed. Specifically, the evaluator first calculates the credit strength among itself, the evaluatee, and guarantors (i.e., neighbors of an evaluator in O , denoted as G) according to their opinions. Then, when the credit strength exceeds a mean value between the subjective consensus level of the evaluator and guarantors, the evaluator will transmit a marked message to evaluatee.

1) **Divergent Level:** As shown in Fig. 2, the evaluator (v_i), guarantor ($v_g \in G$), and evaluatee (v_k) are drawn as the red circle, dark blue star, and purple diamond, respectively. Observably, only one guarantor is drawn to simplify Fig. 2. Then, $a_{i,g} = \{a_{i,g}^1, \dots, a_{i,g}^j, \dots, a_{i,g}^m\}$ represents opinions of v_i on v_g , where m represents the number of attributes of opinions. Similarly, $a_{i,k}$ and $a_{g,k}$ represent opinions of v_i and v_g on v_k .

Based on the above consideration, the difference of opinions between v_i and v_g can be evaluated with the absolute value of multidimensional differences of their opinions. The paradigm of mentioned differences is shown as follows:

$$|a_{i,k} - a_{g,k}| \oplus |a_{i,k} - a_{i,g}|, \quad (14)$$

where $\oplus$ represents joint as shown in Fig. 2. $|\cdot|$ represents the difference of opinions between two vehicles. $|a_{i,k} - a_{g,k}|$ is the dark blue square, and $|a_{i,k} - a_{i,g}|$ is the red square.

Furthermore, the divergent level is calculated according to above differences, and its definition is shown as follows:

Definition 6 Divergent Level: It indicates the mean of opinions' differences of v_i and $v_g \in G$, denoted as $D_{i,g}$.

$$D_{i,g} = \frac{1}{(l+1)m} \sum_j \sum_g^l |2a_{i,k}^j - a_{g,k}^j - a_{i,g}^j|, \quad (15)$$

where l represents the number of guarantors.

The divergent level is a constraint utilized to judge whether their integrated opinions can be acceptable with the association of a dynamical consensus threshold μ (Eq. (13)). If $1 - D_{i,g} \geq \mu$, the divergent level is acceptable.

2) **Credit Strength:** To ensure the reliability of divergent levels, the credit strength is calculated based on opinions of evaluator v_i , guarantor v_g , and evaluatee v_k .

Definition 7 Credit Strength: It indicates the mean of opinions among v_i , $v_g \in G$, and v_k , denoted as $\bar{s}$.

$$\bar{s} = \sum_j^m \sum_g^l \frac{a_{i,k}^j + a_{g,k}^j + a_{i,g}^j}{(2+l)m}, \quad (16)$$

where l represents the number of guarantors, and m represents the number of opinions.

Theoretically, the credit strength is another constraint to evaluate the acceptable degree of integrated opinions, because the higher value of integrated opinions represents credible opinions. Otherwise, in the viewpoint of evaluators, the guarantor cannot provide a guarantee service for an evaluatee.

Based on the definition of credit strength, the larger the value it is, the higher the reliability of this guarantee is, i.e., the marked message³ has a higher probability of being delivered to its destination. Obviously, the credit strength should exceed the average level of opinions of evaluators and guarantors, i.e.,

$$\bar{s} \geq \sum_j^m \sum_g^l \frac{a_{i,g}^j + a_{i,k}^j}{(l+1)m}.$$

F. Dynamic Opinion Mechanism

Based on above considerations, a dynamic opinion mechanism is introduced to adjust opinions of vehicles according to different situations of consensus level similarities. It can be classified into two main situations, i.e., 1) the opinion of a evaluator and its neighbors reach a consensus, and 2) their opinions cannot reach a consensus.

³If a message records a guarantor list, it is named the marked message.

1) **Consensus**: The evaluator (v_i) and its neighbors have a similar view of evaluatee (v_k), i.e., a consensus, which can be divided into positive and negative attitudes. Specifically, the positive attitude represents evaluatee v_k is considered as a benign vehicle, i.e., $(CLS_{i,k} \geq \mu) \wedge (CLS_{o,k} \geq \mu)$. Similarly, the negative attitude represents the evaluatee is considered as a malicious vehicle, i.e., $(CLS_{i,k} < \mu) \wedge (CLS_{o,k} < \mu)$.

Case 1 $((CLS_{i,k} \geq \mu) \wedge (CLS_{o,k} \geq \mu))$: It represents that evaluatee v_k can be employed as a next-hop. Evaluator v_i will transmit a copy of messages to v_k .

Case 2 $((CLS_{i,k} < \mu) \wedge (CLS_{o,k} < \mu))$: It represents that v_k is considered as the malicious vehicle based on opinions of evaluator v_i and guarantors $v_g \in O$. To detect malicious vehicles with collusion behaviors rapidly, v_i decides whether to operate CGM (Section III.E) according to its confidence. Specifically, if the confidence of v_i exceeds the average confidence of its neighbors, v_i will operate the CGM. Otherwise, it does not. Furthermore, the divergent level $D_{i,g}$ and credit strength $\bar{s}$ are calculated, to judge whether a marked message will be transmitted. Specifically, if $D_{i,g}$ and $\bar{s}$ are acceptable, i.e., $1 - D_{i,g} \geq \mu$ and $\bar{s} \geq \sum_j^m \frac{a_{i,g}^j + a_{i,k}^j}{2m}$, v_i will transmit a marked message to v_k .

2) **Non-Consensus**: When the opinion of evaluator v_i and its neighbors $v_g \in O$ cannot reach a consensus, the comprehensive consensus level is calculated to judge whether their opinions are in agreement. Then, v_i makes decisions about whether one more round of opinion adjustment is needed according to the judgment results of consensus levels. Based on this, the non-consensus condition can be divided into two cases, i.e., i) v_i has a positive attitude and its neighbors have a negative attitude $(CLS_{i,k} \geq \mu) \wedge (CLS_{o,k} < \mu)$, and ii) v_i has a negative attitude and its neighbors have a positive attitude $(CLS_{i,k} < \mu) \wedge (CLS_{o,k} \geq \mu)$. The flow of above conditions is shown as Fig. 3.

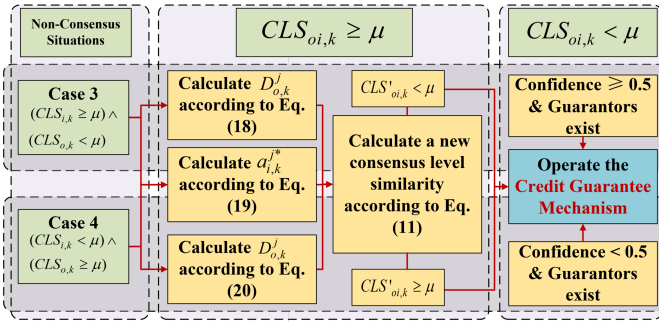


Fig. 3. The flow of non-consensus condition.

Case 3 $(CLS_{i,k} \geq \mu) \wedge (CLS_{o,k} < \mu)$: Evaluator v_i calculates the comprehensive consensus level $CCL_{i,k}$ and its similarity $CLS_{oi,k}$ according to Eq. (7) and Eq. (11). Case 3 is divided into two situations based on $CLS_{oi,k}$ and μ .

- $CLS_{oi,k} \geq \mu$: It represents that v_i has a strong confidence to trust evaluatee v_k . To alleviate messages being delivered to malicious vehicles caused by the overconfidence of v_i , v_i adjusts its opinions toward its neighbors' opinions. Specifically, the j -th opinion with large divergence (denoted as D_{ok}^j) of v_i 's neighbors on v_k should be found

to guide the adjustment of v_i 's opinions, i.e., $1 - D_{ok}^j < \mu$. The divergence of j -th opinion is constructed as follows:

$$D_{ok}^j = \sum_{g \in O} \widehat{w_{g,k}^j} a_{g,k}^j - a_k^j = \sum_{g \in O} \widehat{w_{g,k}^j} (a_{g,k}^j - a_k^j), \quad (17)$$

where $\widehat{w_{g,k}^j}$ represents the normalized weight of j -th opinion, and a_k^j represents the j -th self-certifying opinion. Based on this, if $1 - D_{ok}^j < \mu$, the corresponding opinion should be adjusted as the following rule.

$$a_{i,k}^{j*} = \beta a_{i,k}^j + (1 - \beta) \sum_{g \in O} \widehat{w_{g,k}^j} a_{g,k}^j, \quad (18)$$

where $a_{i,k}^{j*}$ represents the j -th adjusted opinion of v_i . The adjusted opinion is utilized to judge whether an evaluatee can be trusted, i.e., calculating a new consensus level similarity (denoted as $CLS'_{oi,k}$) according to Eq. (11). Then, if $CLS'_{oi,k} \geq \mu$, v_i will transmit messages to v_k . If $CLS'_{oi,k} < \mu$, the CGM will be operated when guarantors exist. Otherwise, v_k is considered as a malicious vehicle.

- $CLS_{oi,k} < \mu$: It represents that evaluator v_i has positive opinions on v_k , but it cannot completely trust evaluatee v_k . To alleviate collusion attacks caused by neighbors, v_i operates the CGM when its confidence exceeds 0.5, because v_i has high confidence and can shoulder the responsibility of failed message transmission. Otherwise, v_i identifies v_k as the malicious vehicle and does not transmit any message.

Case 4 $(CLS_{i,k} < \mu) \wedge (CLS_{o,k} \geq \mu)$: Similar to the process of Case 3, evaluator v_i first calculates the comprehensive consensus level $CCL_{i,k}$ and its similarity $CLS_{oi,k}$. Case 4 is divided into two situations according to $CLS_{oi,k}$ and μ .

- $CLS_{oi,k} \geq \mu$: It represents that v_i cannot complete trust evaluatee v_k , but its neighbors have the positive opinions on v_k . To overcome the prejudice of v_i , v_i considers the opinions of its neighbors comprehensively. Based on this, the j -th opinion with large divergence (denoted as D_{ik}^j) of v_i on v_k should be found by the following rule.

$$D_{ik}^j = |a_{i,k}^j - a_k^j|. \quad (19)$$

If $1 - D_{ik}^j < \mu$, the corresponding opinion should be adjusted according to Eq. (18). Then, the new consensus level similarity $CLS'_{oi,k}$ is calculated according to Eq. (11). If $CLS'_{oi,k} \geq \mu$, the CGM will be employed because guarantors exist and provide positive opinions. Otherwise, v_k is considered as the malicious vehicle.

- $CLS_{oi,k} < \mu$: It involves two situations, i.e., i) although its neighbors have a positive opinion on v_k , v_i with high confidence distrusts v_k . ii) The consensus level similarity between v_i and its neighbors is near to μ with a little fluctuation. To address above situations for decision-making of v_i , the confidence β is utilized to judge whether the CGM can be employed. Specifically, if $\beta \leq 0.5$ and guarantors exist, v_i will generate a marked message to v_k . Otherwise, v_i transmits nothing to v_k .

3) **Opinion Update:** Moreover, the expected opinion will be calculated before message transmission, to update opinions when messages are delivered to their destinations. The tendency of opinion change is different in above situations.

Firstly, for Case 1, the vehicular opinions should be more closely according to their opinions, and its equation is shown as follows:

$$a_{i,k}^{j*} = \beta a_{i,k}^j + (1 - \beta) \sum_{g \in O} \widehat{w_{g,k}^j} a_{g,k}^j. \quad (20)$$

Secondly, for Case 2, evaluator v_i transmits nothing to evaluatee v_k , and the opinion on v_k will not be updated.

Thirdly, for Case 3 and Case 4, the calculation of expected opinions is divided into two situations according to whether the CGM has been employed. When it is not employed, the equation is shown as follows:

$$a_{i,k}^{j*} = \max\{\beta a_{i,k}^j + (1 - \beta) \frac{|\sum_{g \in O} \widehat{w_{g,k}^j} a_{g,k}^j - a_{i,k}^j|}{\max\{\sum_{g \in O} \widehat{w_{g,k}^j} a_{g,k}^j, a_{i,k}^j\}}, a_{i,k}^j\}, \quad (21)$$

where $|\sum_{g \in O} \widehat{w_{g,k}^j} a_{g,k}^j - a_{i,k}^j|$ represents the divergence between the objective and subjective consensus levels. It represents that their comprehensive consensus improves the accuracy of benign vehicle identification, and their opinions should be more consistent.

When the CGM is employed: Guarantor v_g has provided valuable opinions to encourage evaluator v_i in message transmission, so the update rule is related to v_g and is shown as follows:

$$a_{i,k}^{j*} = \beta a_{i,k}^j + (1 - \beta) \sum_{g \in O} \frac{a_{i,g}^j + a_{i,k}^j}{2m}. \quad (22)$$

In addition, when a message is delivered to its destination, the destination checks the truth of received messages and generate a broadcast list (namely delivered message list), to inform vehicles to update their opinions, the weight matrix W_i , and the record of message matrix.

- If the message has been tampered with, opinions of v_i on v_k will decrease according to the following equation:

$$\widetilde{a_{i,k}^j} = \max\{a_{i,k}^j - |a_{i,k}^{j'} - a_{i,k}^{j*}|, 0\}, \quad (23)$$

where $a_{i,k}^j$ represents the opinion of v_i on v_k at present. $a_{i,k}^{j'}$ represents the opinion before the corresponding transmission of messages. $a_{i,k}^{j*}$ represents the expected opinions. Then, the record of message matrix should be adjusted to reduce the weight of v_k and its guarantors v_g . The adjustment equation is shown as follows:

$$d_{i,j}^{t*} = d_{i,j}^t \times \lambda, j \in \{k, g\}, \quad (24)$$

where $d_{i,j}^{t*}$ represents the latest total number of transmitted messages from v_i to $v_j \in \{v_k, v_g\}$, λ represents the punishment strength. Next, the weight of v_i on v_k will decrease Eq. (4).

- If the message has passed verification, opinions increase based on the current opinions and the increment of expected opinions, as shown in follows:

$$\widetilde{a_{i,k}^j} = \min\{a_{i,k}^j + |a_{i,k}^{j'} - a_{i,k}^{j*}|, 1\}, \quad (25)$$

where $|a_{i,k}^{j'} - a_{i,k}^{j*}|$ represents the increment of expected opinions. Furthermore, the number of non-tampered messages ($d_{i,k}^s$) and the total number of transmitted messages will increase ($d_{i,k}^t$), i.e., $d_{i,k}^{s*} = d_{i,k}^s + 1$, $d_{i,k}^{t*} = d_{i,k}^t + 1$. If the CGM is employed, the above values in the message matrix will add one for the guarantors of messages. Moreover, the weight of evaluatee v_k and its guarantors v_g updates according to Eq. (4).

Finally, the result of data delivery affects the confidence of evaluators, and its update equation is shown as follows:

$$\beta = \frac{\psi}{\psi + \phi}, \quad (26)$$

where ψ represents the number of delivered truth messages, and ϕ represents the number of tampered messages.

IV. CONSENSUS-ADAPTIVE TRUST MANAGEMENT

Fig. 4 depicts the flow of CAT. Firstly, vehicles evaluate several consensus levels among their neighbors to judge the consistence of their opinions (Algorithm 1). Secondly, the CGM is introduced to provide a high trustworthiness for message transmission (Algorithm 2). Thirdly, a vehicle classifies its neighbors as several parts with different priorities, to improve the performance of message transmission (Algorithm 3). Finally, the vehicular status is dynamically updated to maintain the precise interaction information (Algorithm 4).

A. Consensus level Evaluation

To precisely evaluate whether a vehicle can be trusted, consensus level evaluation is employed based on the opinion aggregation in groups, especially in the scenario of social relationships. An evaluator calculates several consensus levels according to the opinion and weight of its neighbors, such as the subjective, objective, comprehensive, and self-certifying consensus levels. Furthermore, the illustration of consensus level evaluation is shown in Algorithm 1. The subjective, objective, comprehensive, and self-certifying consensus levels are calculated according to Eq. (5), (6), (7), and (8), respectively.

B. Credit Guarantee Mechanism

The CGM is designed to reduce the influence of untruth opinions provided by malicious vehicles, improving the reliability of message transmission. To try the efforts of evaluators to transmit messages, it filters out a sort of guarantors who provide the confidence and opinions on an evaluatee. Then, the evaluator judges whether the evaluatee gains an opportunity to message transmission according to the divergent level (Eq. (15)) and credit strength (Eq. (16)).

The illustration of CGM is shown in Algorithm 2. First of all, evaluator v_i calculates the divergent level $D_{i,g}$, credit

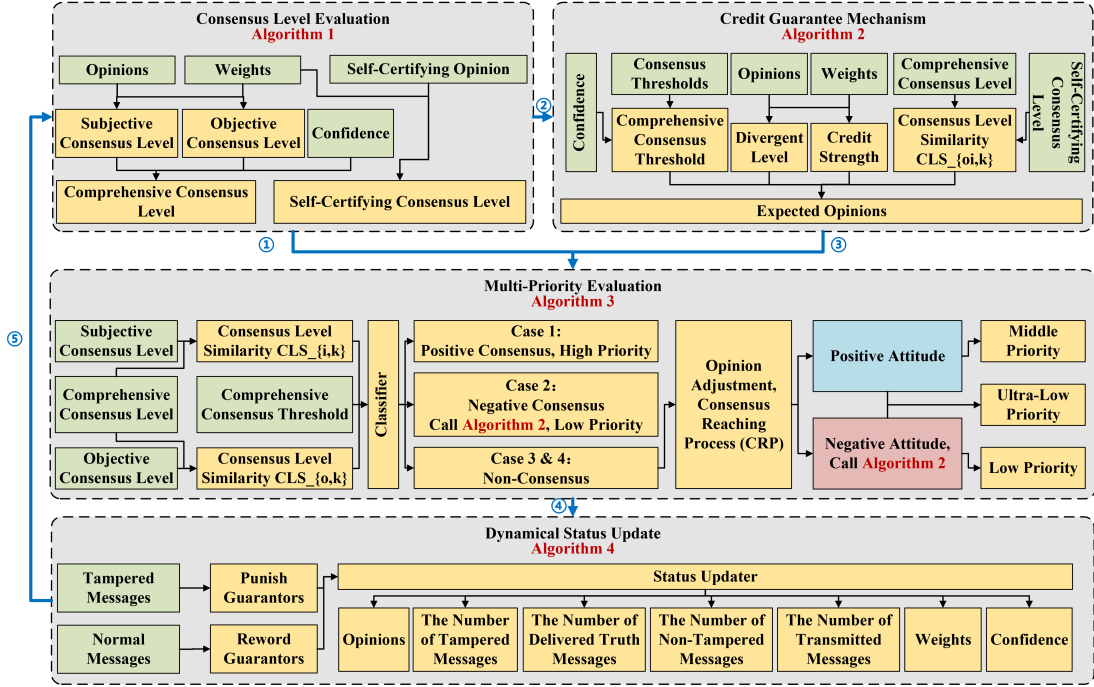


Fig. 4. The flow chart of CAT.

Algorithm 1: Consensus level Evaluation

Input: Neighbors O , Evaluator v_i , Evaluatee v_k

Output: Consensus levels

- 1 Calculate subjective consensus level $SCL_{i,k}$ according to Eq. (5);
- 2 Calculate objective consensus level $OCL_{i,k}$ according to Eq. (6);
- 3 Calculate comprehensive consensus level $CCL_{i,k}$ according to Eq. (7);
- 4 Calculate self-certifying consensus level $SCCL_k$ according to Eq. (8);
- 5 **return** $SCL_{i,k}$, $OCL_{i,k}$, $CCL_{i,k}$, and $SCCL_k$;

strength $\bar{s}$, comprehensive consensus threshold μ , and consensus level similarity $CLS_{oi,k}$ according to Eq. (15), (16), (13), and (11), respectively. Furthermore, to judge whether the guarantee is reliable, v_i should trust guarantors with high weight, and guarantors have positive opinions on evaluatee v_g (i.e., $\bar{s}$ should exceed the average level of opinions of v_i and guarantors, and $CLS_{oi,k}$ should exceed μ). Based on this, if the guarantee is reliable, v_i will calculate the expected opinions according to Eq. (22). The current opinion $[a_{i,k}^1, a_{i,k}^2, \dots, a_{i,k}^m]$ and expected opinions are recorded. Otherwise, the guarantee is unreliable, and v_i will do nothing.

C. Multi-Priority Evaluation

Based on the above consideration, the trustworthiness of an evaluator's neighbors can be evaluated to endow a priority, ensuring the reliability of message transmission. Specifically,

Algorithm 2: Credit Guarantee Mechanism

Input: Guarantors O , Evaluator v_i , Evaluatee v_k

Output: True or False

- 1 Calculate divergent level $D_{i,g}$ according to Eq. (15);
- 2 Calculate credit strength $\bar{s}$ according to Eq. (16);
- 3 Calculate comprehensive consensus threshold μ according to Eq. (13);
- 4 Calculate consensus level similarity $CLS_{oi,k}$ according to Eq. (11);
- 5 **if** $\bar{s} \geq \sum_j^m \sum_g \frac{a_{i,g}^j + a_{i,k}^j}{(l+1)m}$ **and** $CLS_{oi,k} > \mu$ **then**
- 6 Calculate expected opinions $[a_{i,k}^{1*}, a_{i,k}^{2*}, \dots, a_{i,k}^{m*}]$ on v_k according to Eq. (22);
- 7 Record the current opinions and expected opinions;
- 8 **return** True;
- 9 **return** False;

four priorities are considered, i.e., high, middle, low, and ultra-low priorities. Firstly, trusted vehicles are set to a high priority (i.e., Section III.F Case 1). Secondly, if the comprehensive opinion of an evaluator and its neighbors is acceptable after adjusting their opinions, the evaluatee is set to a middle priority. Thirdly, if the CGM is employed according to Algorithm 2, the corresponding vehicle will be set to a low priority. Fourthly, the other vehicles are set to the ultra-low priority.

Algorithm 3 illustrates the process of multi-priority evaluation. It is divided into four cases to empower a priority to evaluatee v_k based on the discussion in Section III.F.

- Lines 1-2: to judge whether the consensus of evaluator v_i and its neighbors is acceptable, consensus level similarities (i.e., $CLS_{i,k}$ and $CLS_{o,k}$) and consensus threshold

Algorithm 3: Multi-Priority Evaluation

Input: Neighbors O , Evaluator v_i , Evaluatee v_k
Output: Multi-priority Queues Q

```

1 Calculate consensus level similarities  $CLS_{i,k}$  and
   $CLS_{o,k}$  according to Eq. (9) and (10), respectively;
2 Calculate consensus threshold  $\mu$  according to Eq. (13);
3 switch  $CLS_{i,k}$  and  $CLS_{o,k}$  do
4   case  $CLS_{i,k} \geq \mu$  and  $CLS_{o,k} \geq \mu$  do //Case 1
5     Calculate and record the expected opinions
      according to Eq. (7);
6     Evaluatee  $v_k$  is set to a high priority;
7   case  $CLS_{i,k} < \mu$  and  $CLS_{o,k} < \mu$  do //Case 2
8     if Algorithm 2 is True then
9       Evaluatee  $v_k$  is set to a low priority;
10  case ( $CLS_{i,k} \geq \mu$  and  $CLS_{o,k} < \mu$ ) or
      ( $CLS_{i,k} < \mu$  and  $CLS_{o,k} \geq \mu$ ) do //Case 3, 4
11    Get  $CCL_{i,k}$  and  $SCCL_k$  from Algorithm 1;
12    Calculate  $CLS_{oi,k}$  according to Eq. (11);
13    if  $CLS_{oi,k} \geq \mu$  then
14      Calculate  $D_{i,g}$  according to Eq. (15);
15      Adjust the corresponding opinions
        according to Eq. (18);
16      Calculate a new consensus level similarity
        ( $CLS'_{oi,k}$ ) according to Eq. (5), (7), and
        (11);
17      if  $CLS'_{oi,k} \geq \mu$  then
18        Calculate and record the expected
          opinions according to Eq. (21);
19        Evaluatee  $v_k$  is set to a middle priority;
20    else
21      if  $CLS_{i,k} \geq \mu$  and  $CLS_{o,k} < \mu$  then
22        if  $\beta \geq 0.5$  and Algorithm 2 is True
          then
23          Evaluatee  $v_k$  is set to a low priority;
24        else
25          if  $\beta < 0.5$  and Algorithm 2 is True
            then
26            Evaluatee  $v_k$  is set to a low priority;
27    otherwise do //other cases
28      if  $v_k$  does not set a priority value then
29        Evaluatee  $v_k$  is set to a ultra-low priority;

```

μ are calculated according to Eq. (9), (10), and (13).

- Lines 4-6: in case 1, v_i and its neighbors consider v_k to be trusted. The expected opinions are calculated according to Eq. (7), and v_k is set to a high priority.
- Lines 7-9: in case 2, v_k is considered an untrustworthy vehicle. To quickly find out other malicious vehicles with collusion behaviors, v_i selects guarantors of v_k and judge whether the CGM can be employed (Algorithm 2). If CGM can be employed, v_k is set to a low priority.
- Lines 10-19: case 3 and case 4 have the same part of processes when the adjusted opinions are acceptable. Specifically, the comprehensive consensus level $CCL_{i,k}$ and self-certifying consensus level $SCCL_k$ are gained from Algorithm 1, to calculate the consensus level simi-

larity $CLS_{oi,k}$ according to Eq. (11). Furthermore, when $CLS_{oi,k} \geq \mu$, it represents that the comprehensive opinions on v_k are positive, thus a new consensus level similarity $CLS'_{oi,k}$ will be calculated. If $CLS'_{oi,k} \geq \mu$, the expected opinions will be calculated and recorded according to Eq. (21). Then, v_k is set to a middle priority.

- Lines 20-26: the judgment condition is different in the last part of case 3 and case 4 based on the confidence of v_i and the result of Algorithm 2. Specifically, in case 3, if the confidence of v_i exceeds 0.5 and the result of Algorithm 2 is true (i.e., the CGM can be employed), v_k is set to a low priority. In case 4, if $\beta < 0.5$ and the result of Algorithm 2 is true, v_k is set to a low priority.
- Lines 27-29: if v_k does not have a priority value, it is set to an ultra-low priority.

D. Dynamical Status Update

Each vehicle updates its status information according to the truth of delivered messages, to make a precise decision based on the newest status information. The status involves the opinions, message matrix, confidence, and weight matrix.

Algorithm 4 illustrates the process of dynamical status update. Firstly, according to the verifying result of messages, vehicles update opinions according to Eq. (23) when the message has been tampered with. Next, the number of tampered messages ϕ will add one. If guarantors exist, they will be punished according to Eq. (24). Secondly, opinions of vehicles will update according to Eq. (25) under the condition that message is benign. The number of delivered truth messages, non-tampered message, and transmitted messages (i.e., ψ , $d_{i,k}^s$, $d_{i,k}^t$) will add one. Next, if the CGM has been employed, the number of non-tampered messages and transmitted messages of guarantors (i.e., $d_{i,g}^s$ and $d_{i,g}^t$) will add one. Thirdly, the weight matrix and confidence update according to Eq. (4) and (26), respectively.

E. Efficiency Analysis

According to the analysis of CRP (Section III.F) and the above flow of algorithms, the time complexity can be analyzed. Specifically, the dimension of equations (e.g., Eq. (5), (6), (7), (8), (13), (15), (16), (21), and (22)) utilized in Algorithms is $|O|$, where $|O|$ represents the number of neighbors, thus the time complexity of them is $O(|O|)$. The time complexity of Eq. (4), (9), (10), (11), (23), (24), (25), and (26) is $O(1)$. Based on this, the time complexity of Algorithm 1, 2, and 3 is $O(|O|)$. The time complexity of Algorithm 4 is $O(1)$. Therefore, the time complexity of CAT is $O(3 \times |O| + 1) = O(n)$.

V. PERFORMANCE EVALUATION

A. Simulation Setup

To evaluate the performance of CAT, the Opportunistic Network Environment (ONE) [38] is introduced with a digital map of Helsinki city, which serves as the base map to support vehicle movement. The simulation time is set to 4,000s for a long-term performance validation relative to several baseline

Algorithm 4: Dynamical Status Update

Input: Delivered Message List M_{ack}

```

1 for  $m_i$  in  $M_{ack}$  do
2   if  $m_i$  is tampered with then
3     Update opinions according to Eq. (23);
4     The number of tampered messages  $\phi \leftarrow \phi + 1$ ;
5     if the CGM has been employed then
6       Punish guarantors according to Eq. (24);
7   else
8     Update opinions according to Eq. (25);
9     The number of delivered truth messages
       $\psi \leftarrow \psi + 1$ ;
10    The number of non-tampered messages
       $d_{i,k}^s \leftarrow d_{i,k}^s + 1$ ;
11    The total number of transmitted messages
       $d_{i,k}^t \leftarrow d_{i,k}^t + 1$ ;
12    if the CGM has been employed then
13      The number of non-tampered messages of
        guarantors  $d_{i,g}^s \leftarrow d_{i,g}^s + 1$ ;
14      The total number of transmitted messages
        of guarantors  $d_{i,g}^t \leftarrow d_{i,g}^t + 1$ ;
15 Update the weight matrix according to Eq. (4);
16 Update the confidence according to Eq. (26);

```

TABLE I
SIMULATION PARAMETERS

Parameters	Values	Parameters	Values
Simulation time	4,000s	The number of vehicles	100
Bandwidths	1Mbit/s	Punishment strength (λ)	1.1
Vehicular velocity	[18 ~ 72]km/h	Message size	1MB
Communication range	200m	The number of message	4500
Attention factor (γ)	0.5	Initial opinions	0.5

algorithms, e.g., DTMS [18], BTCMV [28], and MAGDM [10]. Furthermore, based on the parameters set in [10], [18], [28], [39], 100 vehicles are randomly deployed along the digital map, and their moving speeds are randomly selected from [18 ~ 72]km/h. The communication range, bandwidth, buffer size of vehicles are set to 200m, 1Mbit/s, and 1GB, respectively. Vehicles are divided into benign and malicious vehicles in proportion. Moreover, benign vehicles generate messages with 1MB size, while malicious vehicles operate malicious behaviors. Next, the copies of messages is set to 10 that calculated by 10% of the number of vehicles [40]. Notably, message copies will not affect the network thanks to the following two reasons. 1) Every copy is considered as a new message and will be counted independently. 2) The consensus reaching process being operated before the generation of message copies in the time dimension. In addition, due to the dynamical typology in IoVs, two situations are considered as follows: 1) the new income vehicle who joins a group can provide opinions in the next decision period. 2) the corresponding message will be transmitted to a newly selected next hop when a message transmission interruption occurs (i.e., disconnection). Detailed simulation parameters are shown in Table I.

Based on above settings, the main results of 10 runs show performance with the following indicators:

- **Accuracy:** It represents the percentage of correct identification results and is shown as follows:

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN}, \quad (27)$$

where TP represents the number of true positive⁴, FP represents the number of false positive. Similarly, FN and TN represent the number of false negative and true negative, respectively.

- **Precision:** It represents the accuracy of positive samples and is shown as follows:

$$Precision = \frac{TP}{TP + FP}. \quad (28)$$

- **Recall:** It represents the identification ability of positive samples and is shown as follows:

$$Recall = \frac{TP}{TP + FN}. \quad (29)$$

- **F-score:** It is a measure of predictive performance and is calculated from the precision and recall, as shown as follows:

$$F_\gamma = (1 + \gamma^2) \cdot \frac{Precision \cdot Recall}{(\gamma^2 \cdot Precision) + Recall}, \quad (30)$$

where γ represents the attention factor for precision.

Notably, according to the definition of accuracy, precision, and recall, they are highly related to the ratio of benign vehicles to malicious ones. Although high accuracy and recall are reached in the scenario of a low ratio, low precision also occurs. It is an unexpected result for a system with requirements of high security or robustness. The mentioned three indicators focus on different aspects of identification, but fail to strike a balance among them. Therefore, the F-score is utilized to strike a balance by leveraging the attention of precision and recall.

B. Impact of the number of Malicious Vehicles in CAT

In this section, we consider changing the ratio of benign vehicles to malicious ones from 10% to 90%, denoted as $\xi_{i,j}$, where i and j represent the number of benign and malicious vehicles respectively. Furthermore, the number of messages is employed to explore the influence of message density by changing its generation time, i.e., from 1s to 5s.

1) **The influence of MTA:** Fig. 5a shows that the tendency of accuracy approximately has a downward convex function with the change of $\xi_{i,j}$. The bottom of accuracy occurs between $\xi_{30,70}$ and $\xi_{50,50}$. The lowest value of accuracy is 81.93% in CAT, and the highest value of accuracy is 97.64%. Then, the averages of accuracy with the different numbers of messages (i.e., from 800 to 4,000 messages) are 86.45%, 87.64%, 89.48%, 91.22%, and 93.89%, respectively.

Fig. 5b shows the fluctuation of precision in five scenarios of message densities. All lines have a similar increasing tendency after the first decrease, and their increasing tendencies are

⁴Benign vehicles are correctly identified.

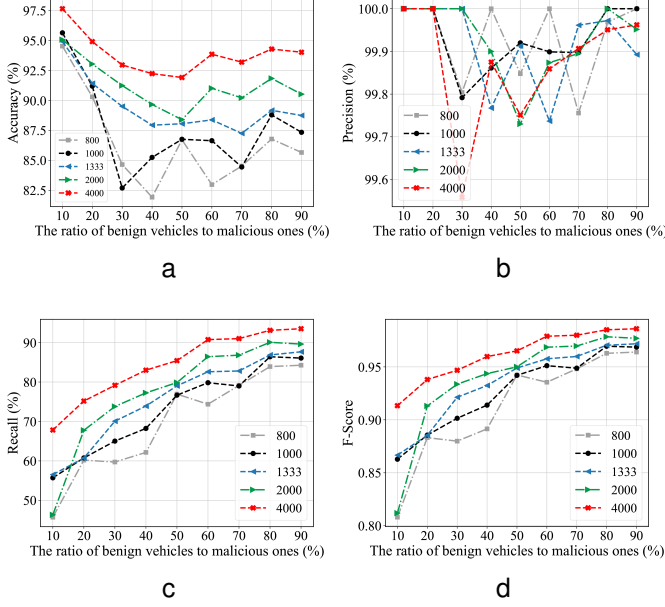


Fig. 5. The performance of CAT with MTA in terms of (a) Accuracy, (b) Precision, (c) Recall, and (d) F-score.

shown beginning at $\xi_{50,50}$. Although the tendency of precision is mutant, the range of variation is actually narrow. For instance, when 4,000 messages exist, the average and Mean Square Error (MSE) are 99.87% and 0.13. Similarly, their values are 99.93% and 0.07 in the scenario of 1,000 messages.

Fig. 5c and Fig. 5d depict the tendency of recall and F-score in variable ratios of benign vehicles to malicious ones, respectively. Specifically, both recall and F-score exhibit a high similarity of increasing tendency with the growth in the number of benign vehicles and messages. When $\xi_{10,90}$ is gradually changed to $\xi_{90,10}$, recall and F-score have the advanced improvement, i.e., 43.28% and 0.17.

Several reasons can illustrate the above phenomena. Firstly, CAT filters out the recommended opinions of untrustworthy vehicles, to prevent it the influence of untruth opinions generated by malicious vehicles. Secondly, several consensus levels are utilized to ensure the consistency of trustworthiness of the evaluatee in groups, including the subjective, objective, comprehensive, and self-certifying consensus levels. Different from traditional ways, the estimation of self-certifying consensus level increases the difficulty of a range of correct opinions guessed by malicious vehicles, because the malicious evaluatee usually cannot know all the encounters of evaluators, their opinions, the weight of opinions, and the comprehensive consensus threshold. Thirdly, the credit guarantee mechanism empowers evaluators to transmit messages to potential benign vehicles according to the confidence of evaluators, improving the identification capability of malicious behaviors.

2) **The influence of HA:** Fig. 6a depicts the tendency of accuracy in the scenario of variable numbers of messages with HA. Specifically, similar to the results of above experiments, a downward convex function is shown in every scenario of message densities. When the number of messages is changed from 800 to 4,000, the averages of accuracy are 86.45%,

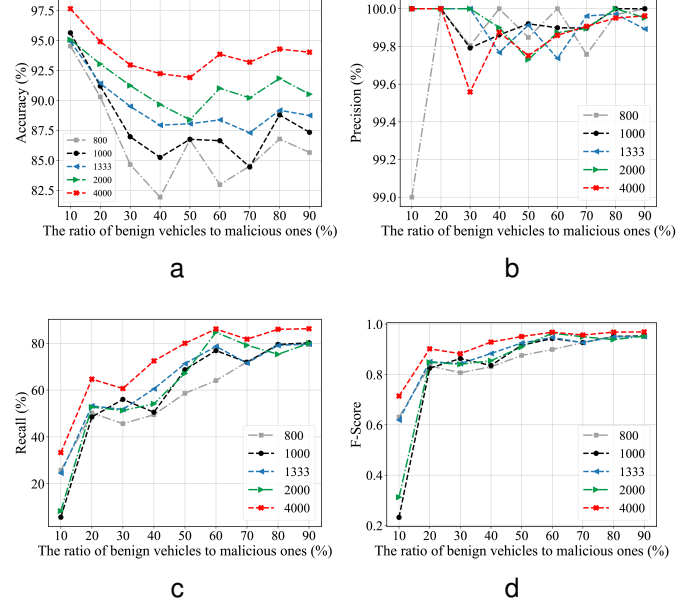


Fig. 6. The performance of CAT with HA in terms of (a) Accuracy, (b) Precision, (c) Recall, and (d) F-score.

88.11%, 89.49%, 91.22%, and 93.89%, respectively.

Fig. 6b shows that CAT performs a stable tendency and an extremely high value of precision. When the number of messages is changed from 800 to 4,000, the averages of precision are 99.82%, 99.93%, 99.92%, 99.93%, and 99.87%, respectively. Similarly, the corresponding values of their MSE are 0.3, 0.07, 0.09, 0.09, and 0.13, respectively.

Fig. 6c shows that recall exhibits an increasing tendency, and the peak is 86.2% when the ratio of benign vehicles to malicious ones is 90%. Then, the increments of recalls are 53.68%, 74.48%, 55.15%, 71.63%, and 52.86%, respectively. The recall in any scenario is over 60% after $\xi_{50,50}$.

Fig. 6d depicts that lines expose the upward concave trend approximately. When the ratio is $\xi_{10,90}$, F-score is the lowest value in five scenarios of different numbers of messages. In the scenario of $\xi_{90,10}$, F-score reaches the top limitation in all lines, i.e., 0.95, 0.95, 0.95, 0.95, and 0.97, respectively.

There are several fundamental reasons to explain the above tendency. Firstly, to adapt to high-risk circumstances (e.g., $\xi_{10,90}$), the untrustworthy opinions will be filtered out. Secondly, thanks to the multi-step evaluation of consensus levels, the process of GDM becomes more reliable based on evaluated consensus levels and their relations. Thirdly, the credit guarantee mechanism expands the interaction of potential trustworthy vehicles, thus messages stored a guarantor list can be utilized to punish and reward guarantors. Based on this, although some guarantors do not receive and relay messages, their guaranteed behaviors will be detected. Then, evaluators can adjust opinions and their weight correctly, according to the guarantor list. Fourthly, the feedback of delivered messages is utilized to update the vehicular status (e.g., opinions, the weight of opinions, and message matrix), thus evaluators can make a precise decision.

Furthermore, the influence of MTA, BMA, CA, and HA is

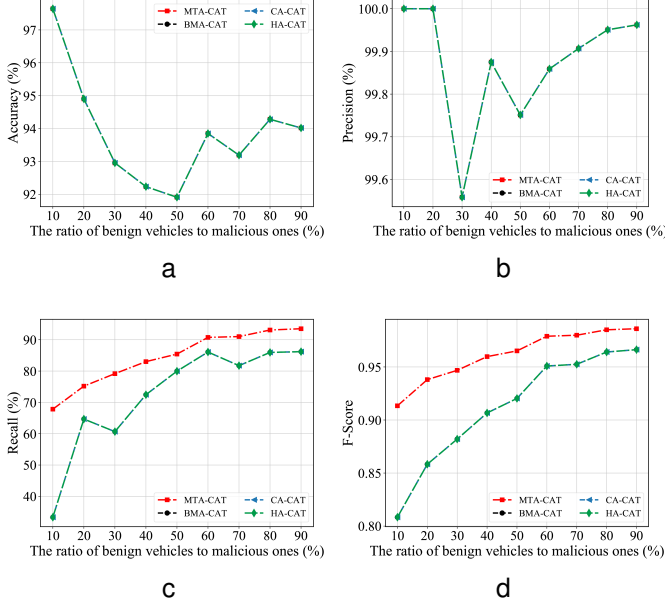


Fig. 7. The performance of CAT with MTA, BMA, CA, and HA in terms of (a) Accuracy, (b) Precision, (c) Recall, and (d) F-score.

explored as shown in Fig. 7 when the number of messages is set to 4000. As shown in Fig. 7a, 7b, 7c, and 7d, BMA, CA, and HA perform the same results in terms of accuracy, precision, recall, and F-score. The cornerstone of the above phenomenon is that both BMA and CA provide false opinions. Specifically, BMA provides a low opinion to benign vehicles, while CA provides a high opinion for malicious vehicles to benign ones. CAT can identify malicious vehicles with CA and BMA accurately, because the false opinions deviate from the right opinions aggregated by evaluators. Moreover, HA consists of BMA and CA, thus it has the same values as BMA and CA.

C. Impact of Various Mechanisms

Based on above experiments, we consider the scenario of 4,000 messages with MTA and HA to show the performance of CAT, BTCMV, DTMS, and MAGDM.

Fig. 8 depicts the performance of CAT, BTCMV, DTMS, and MAGDM in terms of accuracy, precision, recall, and F-score with MTA and HA, respectively. Specifically, Fig. 8a shows various change trends of above four mechanisms in terms of accuracy. CAT and BTCMV have low sensitivity to MTA and HA, while DTMS and MAGDM are sensitive to different attacks. The obvious divergences of accuracy in every mechanism (i.e., CAT, BTCMV, DTMS, and MAGDM) are 0%, 1.85%, 2.7%, and 8.66%, respectively. Furthermore, CAT and BTCMV maintain high accuracy and stability. The averages of accuracy in CAT and BTCMV are 93.89% and 92.66% (MTA), 93.89% and 92.44% (HA), respectively. The trend of DTMS decreases as the number of malicious vehicles decreases, and it converges to 49.18%. MAGDM has an opposite trend to DTMS, and its accuracy is higher than 62% in the scenario of $\xi_{90,10}$.

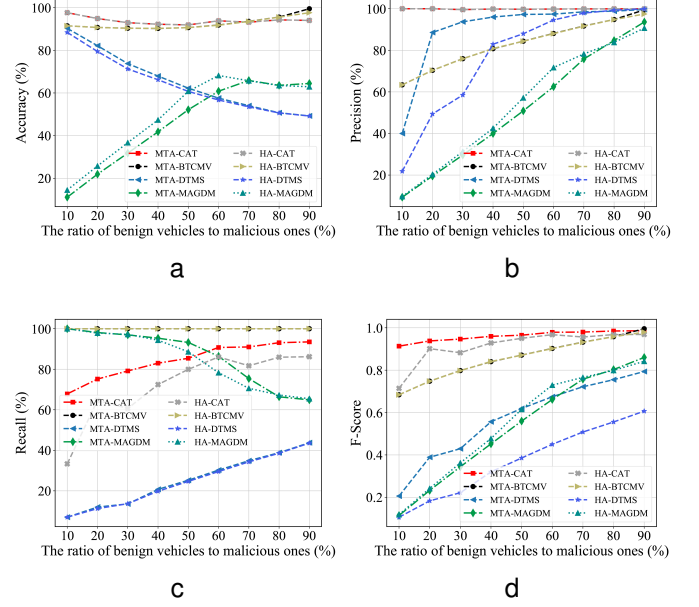


Fig. 8. The performance of four mechanisms in terms of (a) Accuracy, (b) Precision, (c) Recall, and (d) F-score with MTA and HA.

Fig. 8b depicts the increasing tendency of precision in all mechanisms, except CAT. CAT maintains a stable and high performance of precision, i.e., the MSE and average of precision are 0.13 and 99.87% in both MTA and HA. BTCMV, DTMS, and MAGDM are sensitive to $\xi_{i,j}$. When the ratio is $\xi_{10,90}$, their precision is lower than 65%, but rises to 90.69% at least with $\xi_{90,10}$. Furthermore, DTMS and MAGDM are sensitive to both sensitive and different types of malicious behaviors. The largest divergence between the two types of malicious behaviors in DTMS is 44.32%. Similarly, the largest divergence in MAGDM is 9.21%.

Fig. 8c depicts the tendency of recall in CAT, BTCMV, DTMS, and MAGDM with two types of malicious behaviors. Specifically, BTCMV labels all benign vehicles in any scenario, i.e., the recall is 100%. CAT and DTMS have an increasing trend with the increase of $\xi_{i,j}$. MAGDM has an opposite change of the recall curve with other mechanisms. For instance, its value changes from 100% to 65.47% when $\xi_{10,90}$ shifts to $\xi_{90,10}$. Furthermore, BTCMV and DTMS are insensitive to the change in malicious behaviors, while CAT and MAGDM are. With the increase of $\xi_{i,j}$, the influence of MTA and HA decreases in CAT, and MAGDM is easily influenced when $\xi_{i,j}$ is $\xi_{50,50}$ or $\xi_{60,40}$.

Fig. 8d depicts the tendency of F-score with the change of $\xi_{i,j}$ in CAT, BTCMV, DTMS, and MAGDM. Specifically, all of them show a stable upward trend in both MTA and HA, with the increase of $\xi_{i,j}$. CAT maintains a high F-score even in a vulnerable environment (e.g., $\xi_{10,90}$), and outperforms other mechanisms in every scenario. The averages of F-score in CAT are 96% (MTA) and 92% (HA), and the corresponding MSEs are 0.02 and 0.08 respectively. Similarly, BTCMV is slightly affected by the influence of different malicious behaviors, and its F-score is 0.86. Furthermore, both DTMS and MAGDM have a high advancement of F-score with the increase of $\xi_{i,j}$.

and their increments are 0.59 and 0.75 when they undergo MTA. MAGDM is less sensitive to MTA and HA than DTMS. When the ratio is $\xi_{90,10}$, the F-score in DTMS is 0.61 (HA), while the F-score in MTA is 0.79.

Several reasons are listed as follows. Firstly, opinions and their weight in CAT, BTCMV, DTMS, and MAGDM have directly influenced judgments. Specifically, CAT and MAGDM utilize opinions of neighbors (e.g., vehicles) to reach a consensus, but MAGDM fails to eliminate the untrustworthy opinions. Furthermore, both BTCMV and DTMS manage their opinions based on the Bayesian inference theory, but ignore the weight of vehicles.

Secondly, the total number of interactions among vehicles affects accuracy, precision, recall, and F-score. Specifically, CAT and MAGDM do their efforts to reach a consensus by adjusting their opinions, so they have a higher probability of message transmission. CAT filters out untrustworthy opinions to alleviate insidious malicious behaviors, and employs the credit guarantee mechanism to improve the transmission probability, thus CAT adapts to any scenario with high accuracy, precision, recall, and F-score. Furthermore, BTCMV and DTMS contact with trusted vehicles according to a constant trust threshold and trust values, but they are limited by this threshold, reducing the probability of data transmission.

Thirdly, the attention of punishment entails the insidious change of trust evaluation and interferes with the decision of message transmission. Specifically, according to the feedback of delivered messages, CAT punishes vehicles by reducing the weight of opinions and increasing the total number of transmitted messages non-linearly. BTCMV and DTMS reduce trust values to punish vehicles slightly. MAGDM lacks a punishment mechanism.

VI. CONCLUSION

This paper explored the integration of GDM and trust management in IoVs, to adapt to variable environment and improve the identification performance of trust management. A consensus-adaptive trust management was proposed to expand the applied range of trust management and optimize CRP, namely CAT. Specifically, consensus levels were calculated to measure the difference of opinions (trust values) among vehicles. To estimate the reliability of consensus levels, the divergence between consensus levels was calculated, namely consensus level similarity. Furthermore, GDM was introduced to dynamically adjust the opinion of vehicles. Furthermore, a credit guarantee mechanism was designed, to improve the efficiency of CRP and seek out malicious vehicles quickly. Then, to empower the adaptability of trust management for changing environments in IoVs, CAT dynamically managed opinions of vehicles, the self-confidence, and their consensus thresholds according to the feedback of delivered messages. Extensive simulation results showed that CAT can operate in high-risk scenarios, and outperforms other baseline methods in terms of accuracy, precision, recall, and F-score.

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